

OLARINDE OLUBUNMI RACHEAL

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ABOUT ME

Data Scientist/ ML Engineer with end-to-end experience shipping NLP apps (Streamlit + RAG/FAISS) and ethical, user-centered design. Strong in supervised/unsupervised learning, model deployment on AWS, and data storytelling. Passionate about fair, practical AI for healthcare.

TECHNICAL SKILLS

- **Programming:** Python, R, SQL, HTML, CSS
- **Cloud/DevOps:** AWS (S3, SageMaker, Lambda, EC2), Docker (basic), Git, GitHub
- **ML/DS:** Supervised & Unsupervised Learning, Statistical Modeling, Feature Engineering, A/B Testing, EDA
- **Viz & Apps:** pandas, Plotly, Matplotlib, Seaborn, Tableau, Streamlit
- **Deep Learning:** Neural Networks, CNNs, Transfer Learning, Computer Vision, NLP
- **Frameworks:** Scikit-Learn, TensorFlow, Keras, PyTorch, Hugging Face (Transformers), FAISS, OpenCV, spaCy, NLTK, Gensim

EXPERIENCE

Smart Sickle Cell Diary App (Independent)

Jun 2025 - Present

Founder & Lead ML Engineer - Smart Sickle Cell Diary App

- Designed and deployed a Streamlit app (Hugging Face Spaces) for people living with SCD to log symptoms and receive LLM-powered insights with trigger detection (cold, dehydration, stress, exertion).
- Built a lightweight RAG pipeline (FAISS) and Insight Card UX; added tester feedback loop and CSV export for personal records.
- Ensured ethical AI, user agency, and fairness by implementing transparent algorithms and user control features, fostering trust and inclusivity in the app
- Ran a 14-day mini-pilot with de-identified logs; improved onboarding via a how-to video and in-app tips.
- Utilized Python, pandas, FAISS, Streamlit, OpenAI, Hugging Face, and GitHub Pages to develop the app, ensuring robust functionality and seamless integration of technologies
- Developed and launched a portfolio site for the Smart Sickle Cell Diary App using HTML and Streamlit, enhancing user engagement and accessibility

Primrose School at Rancho Sienna, Georgetown TX

Oct 2020 - Feb 2025

Teacher Support

Georgetown TX

- Designed and delivered data-informed lesson plans, applying structured workflows analogous to data pipelines.
- Developed strong communication skills applicable to technical documentation and stakeholder alignment.
- Supported diverse learners, experience that translates to user-centered AI system design

PROJECTS

Smart Sickle Cell Diary App - (Capstone)

- Built an interactive diary app for people living with SCD, combining free-text + light structured inputs and trend visualization for reflection.
- Trained ML models to predict pain intensity (regression) and emotion (classification); exported models with Joblib and integrated into Streamlit.
- Emphasized ethical AI, user agency, and fairness.
- Tools: Streamlit, Scikit-Learn, TextBlob, Plotly, Python, Joblib

Sentiment Analysis of Customer Reviews

- Built and evaluated sentiment classifiers using Word2Vec, Glove, and Sentence Transformers.
- Fine-tuned a transformer from Hugging Face to summarize reviews and extract impactful weekly events.
- Tools: Python, Scikit-Learn, Gensim, Hugging Face Transformers, Matplotlib, Pandas

Landmark Classification Project Using CNNs

- Improved model accuracy from 41% to 79% using ResNet50 with fine-tuning and image augmentation.
- Developed modular PyTorch training scripts and visualized training/validation metrics.
- Tools: PyTorch, OpenCV, Pillow, Livelossplot

Scones Unlimited - ML Workflow with AWS

- Designed a serverless ML pipeline for image classification using AWS SageMaker, Lambda, and S3.
- Deployed and monitored models with SageMaker Model Monitor to detect drift and performance issues.
- Tools: AWS (S3, SageMaker, Lambda), Boto3, Python

EDUCATION

Eastern University, PA <i>Master of Science, Data Science</i> • Achievements: Capstone Projects: “The Model Doesn't Speak My Pain” (algorithmic bias in symptom interpretation)., Smart Sickle Diary App • Coursework: Data Manipulation, Data Science for Business, Data Analytics in R, Data and Database Management with SQL, Cloud Foundations, Fundamentals of Machine Learning Models, Applied Machine Learning, Natural Language Processing, Ethical and Philosophical Issues in Data Science, Applied Data Science & Analytics	Jul 2025
University Of Texas at Austin, TX <i>Postgraduate Certificate, AI/ML</i>	2024 - 2025
Obafemi Awolowo University, Ile-Ife, Nigeria <i>Bachelor of Arts, Music</i>	2011 - 2013
The Polytechnic Ibadan, Ibadan, Nigeria <i>Associate's Degree, Music Technology</i>	2009 - 2010

CERTIFICATIONS/SCHOLARSHIPS

• Udacity Nanodegree Programs:2023 - 2024 AWS Machine Learning Engineer AI Programming with Python ML Fundamentals • Codecademy:2022 - 2024 ML/AI Engineer NLP for Data Scientists Machine Learning for Data Scientists • AWS & Udacity AI/ML Scholar:2023, 2024 Selected among top applicants for hands-on, project-based learning in cloud ML deployment, deep learning, and AI automation.	
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